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**Bridging the Digital Divide: A UX Evaluation of Tech
Accessibility for Older Adults
Undergraduate Thesis**

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ABSTRACT

Digital interfaces are central to how people access services, communicate and manage their lives, yet older adults remain severely underserved by mainstream design. Age-related changes in vision, dexterity and memory can make common interfaces frustrating or inaccessible. This thesis investigates how user experience (UX) design can be improved to reduce cognitive load and usability barriers for adults that are aged 60 and above. Ten participants aged 61–85 from North Texas took part in structured usability sessions followed by semi-structured interviews. Tasks were derived from everyday activities such as finding information on a website, adjusting device settings, composing a message, locating an application and searching for a recipe. Quantitative measures included time on task and success rates, while qualitative data came from participants' comments, pre- and post-task interviews and observational notes.

The findings reveal recurring usability issues that amplify cognitive load for older users: small touch targets and cluttered layouts impede accurate tapping; unclear iconography and nested menus create navigation confusion; websites saturated with pop-ups and adverts distract from primary tasks. Participants with more digital experience completed tasks quickly (median 45–60 s) while novices required several minutes and frequent guidance. Many participants were unaware of built-in accessibility features; those who tried magnifiers or screen readers found them either helpful for reading small text or overwhelming. Themes from the interviews included fear of breaking devices, mistrust of unfamiliar apps, frustration with incessant updates and concerns about online scams. The thesis synthesizes these insights with contemporary research on cognitive aging and accessibility to propose design recommendations centered on simplified navigation, larger icons and fonts, contextual guidance, voice interaction and community-based training. By situating usability as an ethical

imperative rather than a compliance checklist, the study advocates for human-centred design that empowers older adults and reduces their vulnerability to digital exclusion.

CHAPTER 1

INTRODUCTION

1.1 Background and significance

Global demographics are shifting in measurable ways, and the United States reflects this trend clearly. About one in six Americans is now sixty-five or older, and this proportion is expected to keep rising in the coming decades (Ebner & Pehlivanoglu, 2024). This growing population not only manages significant retirement savings but also makes frequent, high-stakes decisions related to health and finances. These factors increase their exposure to exploitation, making older adults prime targets for increasingly sophisticated forms of digital fraud (Ebner & Pehlivanoglu, 2024). In 2021 alone, the Federal Bureau of Investigation documented more than ninety thousand older victims of online scams, with losses surpassing 1.7 billion dollars, representing a seventy-four percent increase in only one year (Ebner & Pehlivanoglu, 2024). While these numbers underscore economic vulnerability, they do not capture the full scope of the digital divide. Many older adults also struggle with basic digital navigation, anxiety about making irreversible errors, and social isolation that further discourages their participation in online environments.

User experience design plays a central role in either reinforcing or alleviating these barriers. Modern interfaces are often developed with younger, digitally fluent users in mind. These users typically benefit from faster processing speeds, refined motor control, and familiarity with shifting design conventions. Although technical standards such as the Web Content Accessibility Guidelines (WCAG 2.2) outline requirements for contrast, text alternatives, and keyboard accessibility (World Wide Web Consortium, 2024), meeting these minimum benchmarks does not guarantee that an interface will feel intuitive or manageable for older adults. Research consistently shows that this group faces challenges related to age-associated declines in vision, fine motor skills, working memory, and attention (Amouzadeh

et al., 2025). When interfaces rely on dense navigation, small touch targets, or rapid task switching, older users often experience frustration, slower task completion, or complete disengagement (Amouzadeh et al., 2025). Because compliance frameworks rarely account for the cognitive workload imposed by real-world interfaces, designers can inadvertently produce systems that are technically compliant yet functionally inaccessible for seniors.

1.2 Problem statement and research questions

This thesis addresses the mismatch between mainstream UX design and the needs of older adults. The central research question is: *How can UX design be improved to reduce cognitive load and usability barriers for older adults when interacting with modern digital technologies?* To answer this overarching question, two supporting questions are explored:

RQ1 – What recurring usability issues do adults aged 60 and above encounter when engaging with commonly used digital interfaces (for example websites, mobile apps and embedded systems)?

RQ2 – Which design strategies and interface adaptations can alleviate cognitive load and enhance the digital experience for older users?

1.3 Scope and definitions

This study focuses on adults aged sixty and older who engage with everyday consumer technologies such as smartphones, tablets, laptops, and web-based applications. The analysis centers on how these individuals navigate routine digital tasks and the challenges that arise from age-related changes in perception, memory, and motor ability. For the purposes of this research, **cognitive load** refers to the mental effort required to interpret information, make choices, and complete actions within a digital interface. High cognitive load can slow task performance, increase errors, and reduce the confidence of older users, especially when interfaces rely on complex workflows or dense visual layouts.

UX design strategies in this context include patterns, interaction models, and interface adjustments intended to support clarity, comfort, and sustained engagement. These strategies aim to reduce friction, promote comprehension, and foster a sense of competence for older adults who may already feel hesitant about digital technologies. The study draws on human-centered and inclusive design principles, viewing accessibility as more than a technical requirement. Instead, accessibility is treated as a commitment to dignity, autonomy, and empowerment. This perspective positions older adults not as passive recipients of assistance but as users who deserve thoughtfully crafted experiences that respect their abilities, preferences, and lived realities.

CHAPTER 2

LITERATURE REVIEW

2.1 Accessibility guidelines and their limitations

The World Wide Web Consortium's Web Content Accessibility Guidelines (WCAG 2.2) provide a widely adopted framework for making digital content accessible to people with disabilities (World Wide Web Consortium, 2024). These guidelines outline practices such as providing text alternatives, ensuring adequate contrast, supporting text resizing, and enabling full keyboard operability across a variety of devices. When implemented, these recommendations tend to improve usability for broad audiences, since they promote clarity, consistency, and predictable interaction patterns (World Wide Web Consortium, 2024).

Despite their value, WCAG 2.2 was not developed to capture the complexity of cognitive aging or the experiential challenges older adults face. Many success criteria within the standard are binary and easily testable, such as verifying the presence of captions or confirming that interface elements meet predefined color contrast thresholds. These criteria do not evaluate whether an interface actually feels intuitive, manageable, or cognitively comfortable for seniors. Scholars have argued that formal compliance often falls short of ensuring meaningful engagement, since the guidelines rarely account for the mental effort required to navigate layered menus, interpret dense information, or manage multitasking within digital systems (Amouzadeh et al., 2025). As a result, a product may meet the letter of accessibility standards yet still place substantial cognitive demands on older adults, leading to frustration, slower task completion, or withdrawal from use.

2.2 Usability barriers for older adults

Empirical research consistently shows that older adults experience difficulty with small touch targets, dense information layouts, and complex navigation structures (Hawthorn, 2007; Charness & Boot, 2009). These issues become more pronounced with age, as declines

in visual acuity, fine motor control, working memory, and attention affect how older users interact with digital systems. A systematic review of mobile app design for older adults found that many products do not meaningfully account for these age-related changes. Instead, developers often rely on design practices optimized for younger populations and rarely include seniors in the design or testing process. This lack of participatory design contributes to poor usability, low adoption, and frequent user frustration (Amouzadeh et al., 2025).

Common barriers include low contrast text, small or tightly spaced buttons, deeply nested menus, and interactions that rely on rapid scrolling or precise gestures. When interfaces fail to accommodate slower information processing speeds or limited dexterity, older adults are more likely to make errors, abandon tasks, or rely on assistance from family members or caregivers. These patterns reflect not a lack of capability, but a mismatch between interface demands and the needs of an aging population.

2.3 Cognitive Aging and Cognitive Load

Research in cognitive aging shows that perceptual decay, reduced attentional inhibition, slower processing speed, and limited working memory resources work together to make complex information more difficult for older adults to manage (Yang & Gan, 2025). These age-related changes affect how efficiently users can filter distractions, maintain focus, and integrate new material while navigating digital systems. Declines in sensory sensitivity and memory also mean that dynamic visual media, including interfaces that rely on motion, animation, or constantly updating elements, impose a heightened cognitive load. Because new information continually replaces what was just presented, older adults must allocate additional effort to keep track of content and maintain orientation within the interface (Yang & Gan, 2025).

These factors complicate tasks such as recalling navigation paths, interpreting unfamiliar icons, or recovering from errors. As cognitive resources are depleted, users may

feel overwhelmed, uncertain, or anxious, which further impairs performance and increases the likelihood of task abandonment. Understanding these cognitive constraints is essential for developing design strategies that promote clarity, reduce mental demand, and support confident interaction for older adults.

2.4 Design strategies and inclusive frameworks

Scholars in human–computer interaction (HCI) have identified a range of design strategies that can better support older adults. Common recommendations include simplified navigation structures, larger and more legible text, voice-activated commands, and interfaces that tolerate errors or provide clearer pathways for recovery (Amouzadeh et al., 2025). These adaptations reduce unnecessary complexity and help compensate for age-related changes in perception, memory, and motor control. Co-designing with older adults has also emerged as a crucial practice. When seniors participate directly in ideation and testing, designers gain insight into needs that may be difficult for users to articulate but critically shape usability and long-term adoption (Amouzadeh et al., 2025).

Research on multimedia learning and information animation further suggests that reducing information complexity, enhancing perceptual cues, and optimising audiovisual integration can improve comprehension, recall, and sustained engagement among older users (Yang & Gan, 2025). These findings highlight the importance of presenting information in ways that support focused attention and reduce cognitive load.

Although these contributions have advanced the field, there is still no universally accepted framework that specifically addresses the cognitive dimensions of aging. Many proposed guidelines remain fragmented or lack rigorous validation across diverse older populations (Yang & Gan, 2025). This gap underscores the need for empirical work that directly links observed usability barriers to systematic and evidence-based design recommendations tailored to cognitive aging.

2.5 Digital vulnerability and ethical considerations

Limited digital literacy significantly heightens older adults' vulnerability to misinformation, phishing, and other online scams. Psychological research indicates that age-related declines in analytical reasoning, working memory, and attentional control reduce the ability to identify deceptive content, including false news articles and fraudulent emails (Ebner & Pehlivanoglu, 2024). This diminished detection capacity has tangible consequences. In 2021, older Americans collectively lost an estimated 1.7 billion dollars to digital fraud, illustrating both the scale and severity of the issue (Ebner & Pehlivanoglu, 2024).

Emotional factors also influence susceptibility. Negative mood, loneliness, and social isolation can weaken judgment and increase compliance with misleading messages (Ebner & Pehlivanoglu, 2024). Scam tactics that rely on reciprocity, such as offering a perceived benefit in exchange for engagement, appear especially effective with older adults, who fall for such strategies more often than scarcity-based approaches (Ebner & Pehlivanoglu, 2024).

These patterns highlight a broader ethical concern. When digital interfaces impose unnecessary cognitive load through cluttered layouts, unclear instructions, or unintuitive workflows, they place older adults at greater risk of misunderstanding content or responding impulsively to deceptive cues. Reducing cognitive load is therefore not only a matter of improving usability. It is a matter of safety, autonomy, and digital well-being for an already vulnerable population.

2.6 Gaps in existing research

Current research often examines accessibility through two separate lenses. One body of work centers on compliance with established accessibility standards, while another focuses on general usability testing. Although these approaches are valuable, they rarely address the specific cognitive changes associated with aging or how these changes interact with everyday

digital tasks. As a result, many studies overlook the nuance required to understand how older adults experience cognitive load in real interfaces.

There is also a methodological gap. Few studies combine quantitative performance measures, such as task accuracy or completion time, with qualitative interviews that explore subjective experiences, frustrations, and decision-making processes. Without this combination, it becomes difficult to map particular design elements to cognitive strain or identify patterns that emerge only when users explain their thought processes.

Additionally, insights from real-world technical support environments are noticeably underrepresented in academic work. Tech support interactions often reveal recurring points of confusion, unanticipated breakdowns, or persistent misconceptions that do not appear in controlled laboratory settings. These observations provide an authentic view of how older adults encounter technology in daily life, yet they are seldom integrated into formal studies.

This thesis aims to address these gaps by collecting a mixed-methods dataset that brings together direct performance metrics, interview narratives, and field-based observations. The goal is to translate these findings into practical design recommendations that support clearer, more intuitive, and cognitively sustainable user experiences for older adults.

CHAPTER 3

METHODOLOGY AND DATA COLLECTION

3.1 Methodology and Data Collection

The study used a mixed-methods research design that combined usability testing, structured observation, and semi-structured interviews. This approach allowed for a detailed examination of how adults aged sixty and older interact with everyday digital technologies and how cognitive load shapes their experiences.

Participants were recruited through social media postings, word-of-mouth referrals, and informal contacts who shared the study with individuals aged sixty and older in the North Texas region. Sessions were conducted individually to reduce pressure and accommodate different comfort levels with technology. Depending on the participant's preference, sessions took place either in their home, in a community setting, or through a remote Zoom meeting. Each participant completed a brief pre-task interview that explored their digital habits, confidence with technology, and any challenges they regularly encountered. This provided contextual information for interpreting their performance during the tasks.

Participants then completed a series of everyday technology tasks using their own devices whenever possible, since familiar hardware reduces anxiety and reflects natural use patterns. When needed, a simplified loaner device was provided. Tasks included searching for information online, navigating a website, sending a message, or adjusting basic settings. During task completion, the researcher observed participants' interactions, noted visible signs of confusion or hesitation, and recorded errors or task breakdowns. With participant consent, sessions were video recorded to support later behavioural analysis.

Following the usability tasks, a post-task interview invited participants to reflect on what felt intuitive, what caused difficulty, and how they approached the tasks. These

reflections provided insight into cognitive effort, emotional responses, and strategies used to cope with unfamiliar or complex interfaces.

The mixed-methods structure provided both quantitative evidence of task performance and qualitative insight into the lived experience of navigating digital interfaces as an older adult.

3.2 Participants

Ten older adults participated in the study, ranging in age from sixty-one to eighty-five years (mean age approximately sixty-nine). The group included six women and four men and reflected a wide range of digital experience. Three participants (P05, P02, and P07) described themselves as very comfortable with technology, four reported feeling somewhat comfortable (P01, P04, P08, and P07), and three indicated low or no comfort with digital devices (P03, P06, P09, and P10).

Participants also varied in the devices they used. Seven owned smartphones or tablets, two relied primarily on laptops, and two used basic flip phones with occasional access to a tablet. These differences helped capture a realistic spectrum of digital familiarity, from daily reliance on online banking and communication tools to minimal engagement due to fear of scams, concerns about breaking devices, or physical limitations such as arthritis.

Table 1 summarises demographic information, self-rated comfort levels, and primary technology habits reported during the pre-task interviews.

Table 1. Participant Demographics

Participant	Age	Gender	Self-rated comfort	Primary devices	Technology habits
P01	61	M	Somewhat	Android phone	daily calls/messages; avoids online shopping; frustrated by small icons
P02	64	F	Very	iPad, laptop	banking and email; recipes on tablet; adjusts settings easily
P03	68	M	Not very	Flip phone, laptop	rare email use; fears breaking devices; relies on family
P04	72	F	Somewhat	Android phone	uses smartphone and desktop; large text and high contrast; avoids new apps
P05	63	M	Very	Windows laptop	part-time work; daily digital tools; uses shortcuts effectively
P06	78	F	Not at all	Basic phone, tablet	landline use; rare digital interaction; fears scams
P07	70	M	Somewhat	Android tablet	reads news and messaging; avoids updates due to layout changes
P08	67	F	Somewhat	iPhone, Chromebook	daily use; frustrated by log-in hurdles; uses voice commands
P09	81	M	Not very	Flip phone, tablet	rare digital use; distrustful of technology; worries about scams
P10	85	F	Not at all	Smartphone rarely, TV/radio	avoids digital tasks; arthritis hampers tapping; uses stylus and large text

3.3 Instruments

The study used three primary instruments: a pre-task interview, a structured set of usability tasks, and a post-task interview. Together, these instruments provided both contextual background and detailed behavioural and experiential data on how older adults navigate everyday digital interactions.

Pre-task interview:

The pre-task questionnaire explored participants' comfort with technology, device habits, and recent experiences with emailing, web searching, or completing routine digital

tasks. It included multiple-choice items, such as self-rated confidence levels and typical device use, along with open-ended prompts about frustrations, avoidance behaviours, and experiences with accessibility features. These responses offered important context for interpreting later task performance and identifying patterns in confidence, hesitation, or reliance on support from others.

Usability tasks:

Participants completed six core tasks modelled on realistic, everyday activities. They were encouraged to use whichever device felt most comfortable, whether a smartphone, tablet, or laptop. The tasks were:

1. Find information on a website:

Participants opened a web browser and located the opening hours of the Denton Public Library. This task assessed navigation strategies, search proficiency, and ability to interpret webpage layouts.

2. Adjust a device setting:

Participants increased text size or screen brightness to evaluate familiarity with settings menus, recognition of icons, and motor precision when interacting with small targets.

3. Send a message:

Using a messaging or email application, participants composed a short message to a dummy contact. They could stop before sending if they preferred. This task measured typing proficiency, awareness of input fields, handling of autocorrect, and comfort with communication apps.

4. Locate and open an app:

Participants found and opened the Calendar application. This tested recognition of icons, familiarity with organisational structures such as home screens or app drawers, and the ability to use device search functions.

5. *Search for a recipe:*

Participants used a search engine to locate a chicken soup recipe. This task explored search literacy, ability to filter results, avoidance of advertisements or misleading links, and judgment of credible sources.

6. *Accessibility feature (optional):*

Participants were invited to activate an accessibility feature such as a magnifier or screen reader. This helped gauge awareness of built-in tools and their initial reactions to assistive options.

Throughout the task session, observations included task completion status (success, partial, or fail), time taken, number and type of errors (such as mis-taps or mis-typings), and any verbal comments relevant to cognitive load or confusion. The researcher offered neutral prompts when participants became stuck but aimed to minimise assistance so natural behaviours could be observed.

Post-task interview:

After the usability tasks, participants answered open-ended questions about their overall experience. They discussed which tasks felt easy or difficult, which design features supported or hindered performance, and whether they felt confident repeating the tasks independently. Participants also reflected on moments of uncertainty, layout or navigation challenges, and daily technology use. These insights contextualised task performance and helped identify patterns related to cognitive load, emotional responses, and user strategies.

3.4 Procedure

Sessions were scheduled at times convenient for participants and typically lasted between forty-five and sixty minutes. After completing the informed consent process, participants engaged in the pre-task interview, which was conducted verbally while the researcher documented responses. Participants were then given a printed list of the usability tasks and encouraged to think aloud as they worked. Timing began when a participant initiated a task and ended when the task was completed, abandoned, or assistance was requested. Once all tasks were finished, or once participants chose to stop, the researcher conducted the post-task interview to gather reflections on their experience. Throughout the session, field notes captured environmental conditions such as interruptions, as well as observable behaviours including hand tremors, visible confusion, moments of enjoyment, and other indicators relevant to cognitive load and user engagement.

3.5 Data Analysis

Quantitative data were analysed using descriptive statistics. Mean and median task completion times were calculated across participants, and success rates and error counts were used to explore differences in performance based on self-reported comfort levels. These measures provided a broad view of efficiency, accuracy, and task difficulty within the sample. Qualitative data from interviews and observational field notes were analysed thematically. Following Braun and Clarke's reflexive thematic analysis approach, the researcher read transcripts multiple times to become familiar with the data, generated initial codes based on recurring ideas such as fear of breaking devices or frustration with advertisements, and then organised these codes into candidate themes. Themes were refined through several iterations and examined in relation to the study's research questions. Representative quotations were selected to illustrate each theme and highlight the lived experience behind the observed behaviours.

CHAPTER 4

FINDINGS

4.1 Task performance overview

Task completion rates were generally high among participants who self-rated as very or somewhat comfortable with technology, but performance declined noticeably among those with low comfort. Table 2 presents a summary of the quantitative performance metrics. The average time to locate the Denton Public Library's opening hours was approximately 2.4 minutes (146 seconds). Participants with higher proficiency completed the task in under a minute by relying on search snippets or built-in shortcuts, whereas novices required up to six minutes and often needed guidance to continue. Adjusting a device setting had a mean completion time of roughly one minute (61 seconds) and was completed successfully by most participants, although some hesitated when navigating nested menus.

Sending a message took an average of 1 minute 54 seconds, but completion times varied widely. Participants unfamiliar with email sometimes typed into the subject line instead of the message body or struggled to locate the send button, requiring correction. Locating the Calendar application was relatively quick, with an average completion time of 35 seconds, particularly when participants used search functions or voice commands. Searching for a recipe produced mixed outcomes. Some participants successfully identified recipes within seconds, while others were slowed by advertisements, pop-ups, or uncertainty about which links were trustworthy. Optional accessibility tasks showed that only a small number of participants were aware of features such as magnifiers, text-to-speech tools, or screen readers, and most encountered these tools for the first time during the session.

Table 2. Task completion times, success rates, and observed issues

Table 2. Task Summary

Task	Average time (s)	Range (s)	Success rate	Observed issues
Find library hours	146	45–360	100 percent with assistance	typing errors; cookie pop-ups; difficulty selecting small menu buttons
Adjust setting	61	10–180	100 percent	confusion between font size and screen zoom; difficulty controlling sliders
Send message	114	30–240	100 percent	autocorrect disruptions; mis-typing; entering text in the wrong field; stylus slips
Open Calendar	35	10–60	100 percent	scrolling across multiple screens; mis-tapping icons; reliance on voice assistant
Search recipe	—	40–120	80 percent	intrusive advertisements; auto-play videos; hesitation selecting a credible source
Use accessibility feature	—	—	60 percent attempted	overwhelming narration from screen reader; confusion when toggling features

Optional tasks were attempted by fewer participants; averages are reported qualitatively.

4.2 Theme 1: Interface complexity and cognitive load

Across participants, small touch targets, cluttered layouts, and deeply nested menus emerged as consistent sources of cognitive load. These design features required precision and continual decision making, which increased hesitation and reduced confidence. P01 explained that he “ended up clicking the wrong thing because the buttons are tiny,” capturing the physical and cognitive strain of interacting with small interface elements. P03 similarly described Windows menus as having “too many options” and reported feeling unsure about where to click. These reactions illustrate how dense visual structures can overwhelm users who already approach technology with caution.

Website interactions presented additional challenges. Cookie banners, drop-down menus, and intrusive advertisements frequently obscured essential information. P02 needed several attempts to locate a “Close” button on a cookie prompt, and P04 observed that the

“Contact Us” drop-down that housed the “Locations and Hours” section on the library website was unresponsive unless tapped with precision. These barriers forced participants to expend extra mental effort on interface management rather than on the task itself, reinforcing a fear of making mistakes and, in some cases, prompting early task abandonment.

Thematic analysis revealed a recurring pattern of “menu overload.” When confronted with long lists of icons or settings, participants struggled to locate the correct path and often described feeling lost. P05 avoided these issues by relying on keyboard shortcuts and search snippets, demonstrating how efficiency increases when interface structures are predictable and easily discoverable. In contrast, P06 and P09 required step-by-step guidance to navigate even basic settings, highlighting how abstracted or hidden functions intensify cognitive load and impede independent exploration.

4.3 Theme 2: Input challenges and motor limitations

Input-related difficulties were common across participants and contributed significantly to task breakdowns. Typing on small on-screen keyboards proved especially challenging. Autocorrect routinely introduced errors that disrupted the flow of typing and increased cognitive load. For example, “usability” became “visibility” for P01, and P04 watched “library” transform into “lobotomy,” prompting both to backtrack and correct text multiple times. These interruptions not only slowed task completion but also heightened frustration and uncertainty about whether the device was responding as intended.

Motor limitations further amplified these challenges. Participants with arthritis or hand tremors, such as P10, struggled to tap accurately. Her stylus slipped during the messaging task and deleted text she had already composed, leading to visible frustration. Environmental distractions also played an unexpected role. P02’s cat walking across her lap and P08 responding to a doorbell disrupted their concentration, demonstrating how divided attention can make small input errors more frequent and increase cognitive effort.

Participants who used voice dictation appreciated its convenience but encountered new types of errors. Misinterpretations were common, such as P06's attempt to say "usability study," which the device transcribed as "use a billy stew." These errors required additional correction that negated some of the intended efficiency of dictation.

Motor challenges also affected interactions beyond typing. Slider controls, such as those used to adjust screen brightness, required fine motor precision that some participants lacked. Several needed repeated attempts to position the slider correctly. These issues illustrate how even simple actions can become taxing when physical limitations interact with interface designs that demand precision.

4.4 Theme 3: Navigation strategies and mental models

Participants used a wide range of strategies to locate applications and move through interface environments, revealing clear differences in navigation efficiency and underlying mental models. More experienced users, such as P05 and P07, relied on search functions or keyboard shortcuts and completed navigation tasks quickly. Their actions suggested well-developed mental models that matched the organisational logic of their devices. P04 demonstrated the value of multimodal interaction by using voice commands, saying "Open my calendar," which allowed her to bypass visual search entirely.

In contrast, several participants navigated by scrolling through long lists of icons or screens. P03 and P06 frequently mis-tapped adjacent icons, such as opening Calculator instead of Calendar, which indicated both motor strain and uncertainty about how apps were arranged. These behaviours pointed to a mismatch between their expectations of where functions should be located and the actual structure of the interface. As a result, many relied on trial-and-error exploration, a strategy that increased cognitive load and often led to visible anxiety.

Participant comments underscored this tension. P03 remarked, “I would have given up at home,” signalling that unsupported exploration felt overwhelming outside the controlled study setting. Similarly, P09 asked, “Why do they hide the simple stuff?” expressing frustration with functions that felt buried or poorly labelled. These experiences highlight how unclear interface organisation disrupts older adults’ ability to form reliable mental models, making routine navigation slower, more stressful, and less intuitive.

4.5 Theme 4: Awareness and perception of accessibility features

Awareness of built-in accessibility features was generally low. Only three participants (P02, P04, and P07) reported prior experience using tools such as larger text, high contrast settings, or magnifiers. The remaining participants were unfamiliar with features designed to support visual or motor challenges, and several expressed surprise that such options existed. When prompted to explore them during the optional task, reactions varied widely.

Some participants responded positively. P04 praised the screen magnifier for making small text easier to read, and P07 commented that he “might use it now and then” for detailed content. P02 tried the Speak Screen function on her iPad and initially laughed at the unexpected narration, later acknowledging that it could be useful when reading long articles or recipes. These responses reflect a general openness to tools that reduce visual strain or streamline information processing.

Other participants found accessibility features confusing or overwhelming. P06 and P08 both turned off the screen reader almost immediately, describing the rapid narration as disorienting. P09 appreciated the magnifier for crossword puzzles but forgot to disable it, resulting in a zoomed-in interface that left him momentarily confused about how to return to the normal view. These experiences demonstrated that unfamiliar or intrusive accessibility tools can increase cognitive load rather than reduce it.

Overall, the findings suggest that discoverability and user control are essential. Accessibility features need to be easy to activate and deactivate, with clear explanations of what they do and when they might be helpful. Without intuitive entry and exit points, even well-designed features risk becoming barriers rather than supports.

4.6 Theme 5: Emotional responses and attitudes toward technology

Participant experiences extended beyond measurable performance outcomes and revealed a broad emotional landscape that shaped how older adults approached digital tasks. Fear of “breaking” devices was especially common among less confident users such as P03, P06, P09, and P10. This anxiety led to slow, tentative interactions and reluctance to explore unfamiliar icons, menus, or settings. For many, the risk of making an irreversible mistake felt greater than the potential benefit of trying something new.

Several participants framed technology use as a personal cost–benefit calculation. P09 expressed limited motivation to learn new systems, stating, “At my age, I’d rather garden than learn these things,” while P10 felt that constant software updates made learning new features seem futile. These perspectives highlight the role of perceived relevance. If the value of digital engagement is unclear, older adults may opt out entirely, reinforcing the digital divide.

However, not all emotional responses were negative. Participants such as P05 and P02 approached the tasks with curiosity and enjoyment, expressing interest in features they had not previously explored. Their engagement suggested that a sense of mastery, or even small moments of successful interaction, can foster positive attitudes toward technology and encourage continued use.

Together, these findings show that emotional factors, including confidence, fear, frustration, and curiosity, are central to understanding how older adults navigate digital

environments. Designing for this population therefore requires attention not only to cognitive and physical demands but also to the affective dimensions of technology use.

4.7 Theme 6: External factors and vulnerability

Concerns about scams, misinformation, and privacy shaped participants' willingness to engage with digital tasks. Several participants described a general sense of risk associated with online environments. P04 avoided downloading new applications because she feared they might contain malware, while P06 and P09 expressed worry about being scammed and therefore refrained from using online banking altogether. These concerns mirror broader evidence that older adults are disproportionately targeted by financial exploitation and that age-related declines in analytical reasoning and memory can increase susceptibility to phishing attempts (Ebner & Pehlivanoglu, 2024). Fear of harm therefore played a significant role in shaping digital behaviour.

Some participants managed this vulnerability by outsourcing digital tasks to family members. While this strategy reduced immediate risk, it also reinforced long-term dependency and limited opportunities to develop confidence. This pattern reflects a broader cycle in which fear-driven avoidance restricts digital engagement, which in turn maintains low competence and heightens vulnerability.

Environmental factors also influenced performance. Interruptions such as children passing through the room, pets seeking attention, or unexpected doorbells disrupted concentration and increased error rates. These real-world distractions revealed how easily cognitive load can escalate when attention is divided, highlighting the need for interfaces that can tolerate pauses, support task recovery, and remain clear even under less-than-ideal conditions.

Overall, the theme underscores that vulnerability is shaped not only by interface design and cognitive load but also by emotional, social, and environmental contexts.

Effective solutions must therefore address both individual concerns and the broader circumstances in which older adults use technology.

CHAPTER 5

DISCUSSION

5.1 Interpreting the findings through cognitive aging theory

The usability barriers identified in this study can be understood through established principles of cognitive aging. Participants who struggled with small touch targets, cluttered layouts, and rapidly shifting menus were likely experiencing reduced perceptual acuity and slower processing speed, both of which are well documented in cognitive aging research (Yang & Gan, 2025). These limitations made precise interactions more demanding and contributed to hesitation and error, particularly when tasks required quick visual discrimination.

The cognitive load imposed by dynamic websites, intrusive pop-ups, and advertisements reflects findings that continuous information replacement places substantial demands on working memory (Yang & Gan, 2025). Several participants were forced to reorient themselves repeatedly as new elements appeared on screen, increasing mental effort and reducing task efficiency. This pattern aligns with research showing that older adults have fewer cognitive resources available for managing multiple streams of visual information, which can lead to confusion and task abandonment.

Conversely, participants who used voice commands or keyboard shortcuts demonstrated strategies that distributed cognitive effort across different modalities. By shifting part of the workload away from visual scanning or fine motor control, these users reduced their reliance on limited cognitive resources. This aligns with multimedia cognitive load theory, which suggests that combining visual and auditory channels can enhance performance when the integration is intuitive and does not overwhelm the user (Yang & Gan, 2025). These findings illustrate how supportive interaction modalities can compensate for age-related cognitive changes and promote more confident engagement with technology.

5.2 Recurring usability issues (RQ1)

The first research question examined the recurring usability issues experienced by older adults during everyday digital tasks. Several consistent barriers emerged across participants.

Small touch targets and low-contrast text:

Buttons, icons, and menu links often required precise tapping, creating both cognitive and motor strain. Participants reported that larger targets and high-contrast modes made interactions more manageable.

Complex navigation and hidden menus:

Long settings lists, deeply nested drop-downs, and ambiguous icons contributed to disorientation. Users expressed a preference for clear labels, predictable layouts, and direct access to commonly used functions.

Distracting advertisements and pop-ups:

Websites cluttered with cookie notices, advertisements, and auto-play videos diverted attention from the primary task. These interruptions increased cognitive load and frequently forced participants to pause, reorient, or seek assistance.

Input challenges:

On-screen keyboards and autocorrect errors introduced friction, especially for participants with motor limitations. Stylus use reduced finger strain for some but introduced its own difficulties, such as accidental dragging or unintended text deletion.

Limited awareness of accessibility features:

Most participants were unfamiliar with built-in accessibility tools, including magnifiers, text-to-speech, or screen readers. Those who experimented with these features recognised their potential benefits but emphasised the need for easy toggling and clear explanations of their purpose.

Emotional factors:

Fear of making mistakes, frustration with frequent software updates, and low perceived relevance shaped engagement. Participants with more positive attitudes or a sense of curiosity approached tasks with greater confidence and navigated complexity more effectively.

Together, these findings reveal a multilayered set of barriers that reflect both interface design limitations and the broader cognitive, motor, and emotional contexts in which older adults use technology.

5.3 DESIGN STRATEGIES TO REDUCE COGNITIVE LOAD (RQ2)

The second research question focused on identifying design strategies that can reduce cognitive load and improve usability for older adults. The findings from this study point to several principles that can inform more inclusive and supportive interface design.

Prioritise clarity and simplicity:

Interfaces should minimise visual clutter, use high contrast, and rely on large, clearly labelled buttons to reduce perceptual strain. Progressive disclosure can help manage complexity by presenting essential options first and revealing advanced settings only when needed. This approach reduces menu overload and supports users who may be hesitant to explore densely nested structures.

Optimise navigation structures:

Shallow hierarchies and consistent placement of controls help older users form reliable mental models. Navigation aids such as breadcrumbs or contextual cues can convey a user's current position within an app or website. Critical functions should not be hidden behind small drop-downs or ambiguous icons, which were frequent sources of confusion in the study.

Support multimodal input:

Alternative input methods, including voice commands, search bars, and hardware shortcuts, can compensate for visual or motor limitations. These tools distribute cognitive effort more evenly across modalities. However, they should remain optional and accompanied by brief guidance, since unfamiliar input modes can otherwise introduce new sources of confusion.

Enhance discoverability of accessibility features:

Participants' limited awareness of built-in accessibility tools suggests a need for clearer onboarding. Tutorial prompts or short explanations can introduce features such as magnifiers, high-contrast modes, or screen readers. These tools should be easy to toggle on and off, with one-step activation and plain-language descriptions of their benefits.

Reduce external distractions:

Websites designed for older audiences should limit advertisements, avoid auto-play media, and prevent pop-ups from obscuring core content. When ads must be included, they should be placed in non-intrusive locations and not require precise tapping to dismiss.

Foster trust and safety:

Interfaces should communicate security clearly through indicators of encrypted connections, straightforward explanations of permission requests, and built-in resources that educate users about common scams. Designers should avoid dark patterns and should not replicate the persuasive tactics often used by fraudsters.

Co-design with older adults:

Involving seniors throughout the design process ensures that products align with real-world needs and cognitive frameworks. Participatory methods, including workshops and iterative testing, can reveal barriers that may not surface through heuristic evaluation alone and can validate proposed solutions (Amouzadeh et al., 2025).

Taken together, these strategies offer practical pathways for reducing cognitive load and enhancing digital accessibility, supporting older adults in navigating technology with greater confidence and autonomy.

5.4 Ethical implications

The emotional responses observed in this study, including fear, frustration, and resignation, underscore that accessibility is not solely a technical challenge but also an ethical one. As the proportion of older adults continues to grow, inadequate digital design has the potential to deepen social isolation, reduce autonomy, and increase susceptibility to harm. Psychological research shows that age-related declines in analytical reasoning and memory can reduce the ability of older adults to detect deception, making them more vulnerable to misinformation and scams (Ebner & Pehlivanoglu, 2024). When digital interfaces are overly complex, ambiguous, or manipulative, they may inadvertently resemble the persuasive tactics used by fraudsters, thereby intensifying risk.

Ethical design therefore requires more than basic compliance with accessibility guidelines. It calls for an empathetic understanding of the cognitive and emotional landscapes of older users. Designers must remain aware of the heightened vulnerability within this population and avoid patterns that could be misinterpreted, misleading, or cognitively taxing. Clear communication, transparent information flows, and predictable interaction pathways can help promote digital safety and autonomy. Ultimately, treating accessibility as a moral responsibility ensures that older adults are supported rather than excluded as digital systems become increasingly integral to everyday life.

5.5 Comparison with prior research

The findings of this study corroborate earlier research showing that older adults frequently encounter difficulties with small touch targets, complex layouts, and inaccessible settings (Hawthorn, 2007; Fisk et al., 2009). By combining qualitative narratives with

quantitative performance data, the study also extends existing work by illustrating how these barriers manifest in real time and how they interact with emotional and contextual factors. While the WCAG provides broad guidance for improving accessibility (World Wide Web Consortium, 2024), the results reveal that cognitive load can remain high even when interfaces meet technical requirements. This supports critiques suggesting that compliance alone does not guarantee usable or comfortable interaction for older adults.

The themes identified here align with systematic reviews that advocate simplified navigation structures and co-design with older adults to ensure that products reflect their needs and mental models (Amouzadeh et al., 2025). They also support cognitive aging theory, which links sensory decline and slower processing speed to increased difficulty managing complex or dynamic information (Yang & Gan, 2025). Furthermore, the emphasis on emotional responses and trust broadens the discourse by highlighting the ethical dimensions of technology design. These findings resonate with research documenting heightened vulnerability to scams and misinformation among older adults (Ebner & Pehlivanoglu, 2024), suggesting that usability and online safety are closely intertwined.

5.6 Limitations and future research

This study has several limitations that should be considered when interpreting its findings. The sample size was small and drawn exclusively from the North Texas region, which limits the generalisability of the results. Participants also self-selected into the study, which may have excluded individuals who are entirely disconnected from technology or who feel too intimidated to participate in research involving digital tasks. Furthermore, the predetermined task set, while grounded in everyday activities, may not fully capture the breadth of digital interactions that older adults encounter in their daily lives.

Future research would benefit from larger and more diverse samples, including participants with varying levels of digital disengagement. Longitudinal studies could provide

insight into how training programs, interface redesigns, or gradual exposure to new technologies influence confidence and skill development over time. Examining the role of cultural background, socioeconomic status, and social support networks may also yield richer understanding of how structural factors shape digital engagement.

There is substantial opportunity for experimental testing of specific design changes. For example, measuring the effects of increased button size, simplified menu structures, or alternative input modalities on performance and cognitive load could offer evidence-based guidelines for inclusive design. Finally, emerging technologies such as machine learning present possibilities for adaptive interfaces that respond dynamically to user behaviour. While promising, such approaches raise ethical questions related to data privacy, autonomy, and the risk of over-personalisation, all of which warrant careful examination in future work.

CHAPTER 6

CONCLUSION AND RECOMMENDATION

This thesis set out to examine how UX design can reduce cognitive load and address usability barriers experienced by older adults. Through a mixed-methods study involving ten participants, the research identified a set of recurring challenges and articulated design strategies grounded in both empirical observation and established literature. Small touch targets, complex navigation structures, distracting visual content, input challenges, and limited awareness of accessibility features emerged as central obstacles. Emotional responses such as fear of mistakes, frustration with software changes, and low motivation further influenced engagement and shaped participants' willingness to explore unfamiliar features. At the same time, the study revealed resilience and adaptability. Participants who used shortcuts, voice commands, or accessibility tools demonstrated higher confidence and more successful task completion.

To help bridge the digital divide, designers and policymakers must adopt a holistic understanding of accessibility that goes beyond technical compliance. Simplifying interfaces, improving the discoverability of assistive features, supporting multimodal input, and involving older adults throughout the design process can meaningfully reduce cognitive load and enhance user confidence. Educational efforts should combine digital literacy support with practical guidance on identifying scams and deceptive content, addressing both technical and psychological vulnerabilities that increase risk. These interventions recognise that technology use among older adults is shaped not only by cognitive and motor changes but also by trust, motivation, and perceived relevance.

Ultimately, inclusive UX design is a commitment to preserving independence and dignity for older adults. Thoughtfully designed technologies can empower users to engage more fully in digital life, remain connected to essential services, and avoid forms of

exploitation that disproportionately affect aging populations. By centring empathy and recognising the diverse needs of older adults, designers can create digital environments that truly serve all members of society.

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